ABG, ECG, Heart Block & Pacing – Chapter Summary

Systematic Interpretation • Conduction • Pacing Modes

ICU / EDIC Board Review • MCQ-Oriented

1. ABG Step-by-Step

- **1. Validity:** H^+ (nmol/L) ≈ 24 × $PaCO_2$ / HCO_3
- **2. Acidemia / alkalemia:** pH < 7.35 / > 7.45
- **3. Primary:** $PaCO_2$ (resp) vs HCO_3 (metabolic)
- **4. Compensation** appropriate?
- **5. Anion gap** (corrected for albumin)
- **6. Delta ratio** if ↑ AG
- **7. Oxygenation:** A-a gradient, P/F

Normal	Value
pH	7.35–7.45
$PaCO_2$	35–45 mmHg
HCO_3	22–26 mmol/L
BE	−2 to +2
AG	8–12 (no K)
Lactate	< 2 mmol/L

2. Compensation Rules

Disorder	Expected
Metabolic acidosis	$PaCO_2$ = 1.5 × HCO_3 + 8 ± 2 (≈ last 2 digits of pH)
Metabolic alkalosis	$PaCO_2$ = 0.7 × HCO_3 + 21 ± 2
Acute resp acidosis	HCO_3 ↑ 1 per 10; pH ↓ 0.08
Chronic resp acidosis	HCO_3 ↑ 3.5–4 per 10; pH ↓ 0.03
Acute resp alkalosis	HCO_3 ↓ 2 per 10
Chronic resp alkalosis	HCO_3 ↓ 4–5 per 10

- Compensation never overcorrects pH
- Normal pH with abnormal $PaCO_2$ & HCO_3 = mixed disorder

3. High AG Metabolic Acidosis

- **GOLDMARK:** Glycols (ethylene, propylene), Oxoproline (paracetamol), L-lactate, D-lactate, Methanol, Aspirin, Renal failure, Ketoacidosis
- **Osmolar gap** = measured − (2Na + glucose/18 + BUN/2.8); > 10 → toxic alcohol
- **Ethylene glycol:** oxalate crystals, AKI, hypocalcemia
- **Methanol:** visual loss, basal ganglia
- **Propylene glycol:** lorazepam / diazepam infusions
- **D-lactate:** short bowel syndrome
- **Pyroglutamic acidosis:** chronic paracetamol + malnutrition, sepsis

4. Delta Ratio & Normal AG

- **Δ ratio** = (AG − 12) / (24 − HCO_3)

Δ ratio	Meaning
< 0.4	Pure NAGMA
0.4–0.8	Combined HAGMA + NAGMA
1–2	Pure HAGMA (DKA ~1, lactic ~1.6)
> 2	HAGMA + metabolic alkalosis / chronic resp acidosis

- **NAGMA "HARDASS":** Hyperalimentation, Acetazolamide, RTA, Diarrhea, Addison's, Spironolactone, Saline
- **Urine AG** negative = GI; positive = renal (RTA)

5. Metabolic Alkalosis

- **Chloride-responsive** (urine Cl < 20): vomiting, NG suction, diuretics (late), post-hypercapnia → saline + KCl
- **Chloride-resistant** (urine Cl > 20): hyperaldosteronism, Cushing, Bartter / Gitelman, severe hypoK → treat cause, spironolactone
- **Acetazolamide** for alkalosis in ventilated / HF patients
- Severe (pH > 7.6): HCl, dialysis

6. ECG Fundamentals

Interval	Normal
PR	120–200 ms
QRS	< 120 ms
QTc (Bazett)	< 450 ♂ / < 460 ♀; > 500 = high risk

- **Axis:** LAD (LAFB, inferior MI, LVH), RAD (RVH, PE, LPFB, lateral MI)
- **RBBB:** RSR' in V1, wide S in I, V6
- **LBBB:** broad R in I, V6; deep S in V1; new LBBB + ischemia → Sgarbossa criteria
- **LVH:** Sokolow–Lyon S(V1) + R(V5/6) > 35 mm
- **PE:** sinus tachycardia (commonest), S1Q3T3, RBBB, right-axis, T inversion V1–V4

7. ST Changes & Ischemia

- **STEMI:** ST ↑ ≥ 1 mm in 2 contiguous leads (V2–V3: ≥ 2 mm ♂ > 40, ≥ 1.5 mm ♀)

Territory	Leads	Artery
Inferior	II, III, aVF	RCA (80%) / LCx
Anterior	V1–V4	LAD
Lateral	I, aVL, V5–V6	LCx / diagonal
Posterior	ST ↓ V1–V3, tall R	RCA / LCx → V7–V9
RV	ST ↑ V4R	Proximal RCA

- **STEMI equivalents:** de Winter T waves, Wellens (LAD), posterior MI, aVR elevation + diffuse ST ↓ (LMCA / 3-vessel)
- **Pericarditis:** diffuse concave ST ↑ + PR depression (PR ↑ in aVR)

8. Electrolyte & Drug ECG Signs

Condition	ECG
Hyperkalemia	Peaked T, wide QRS, sine wave
Hypokalemia	U waves, ST ↓, flat T
Hypercalcemia	Short QT
Hypocalcemia	Long QT
Hypothermia	Osborn (J) waves, bradycardia
Digoxin	Scooped ST, bidirectional VT, PAT with block
TCA / Na-channel blockers	Wide QRS, R in aVR > 3 mm → $NaHCO_3$
Brugada	Coved ST ↑ V1–V2
Hypomagnesemia	Torsades

9. Heart Blocks

Block	ECG	Site	Rx
1st degree	PR > 200	AV node	None
Mobitz I (Wenckebach)	Progressive PR ↑ → dropped QRS	AV node	Usually benign
Mobitz II	Fixed PR, sudden dropped QRS	His–Purkinje	Pacemaker
2:1	Every other P	Node or below	Narrow QRS → node; wide → infranodal
3rd degree	AV dissociation, P–P & R–R regular	Node (narrow escape) or below (wide)	Pacing

- **Inferior MI:** AV-nodal block (transient, atropine-responsive)
- **Anterior MI:** infranodal block (poor prognosis, needs pacing)
- **Bifascicular** = RBBB + LAFB/LPFB; **trifascicular** + 1st-degree block

10. Bradycardia Algorithm

- **Adverse features:** shock, syncope, myocardial ischemia, heart failure
- **1. Atropine 1 mg IV** (repeat to 3 mg) – ineffective in infranodal block & heart transplant
- **2. Isoprenaline / adrenaline 2–10 µg/min / dopamine**
- **3. Transcutaneous pacing** → transvenous pacing
- **Risk of asystole:** recent asystole, Mobitz II, complete heart block with wide QRS, ventricular pause > 3 s
- **Causes:** ischemia, drugs (β-blockers, CCB, digoxin), hyperK, hypothermia, hypothyroid, ↑ ICP (Cushing)
- β-blocker OD → glucagon, high-dose insulin; CCB OD → calcium, HIET, lipid

11. Temporary Pacing

- **Transcutaneous:** pads anterior–posterior; ↑ mA until capture (50–100 mA), sedation; confirm with pulse (not ECG)
- **Transvenous:** right IJ or left subclavian; RV apex; threshold < 1 mA; set output 2–3 × threshold
- **Epicardial wires** after cardiac surgery
- **Complications:** pneumothorax, perforation / tamponade, infection, arrhythmias, failure to capture / sense

12. Pacing Codes (NBG)

Position	Meaning	Letters
I	Chamber paced	A, V, D, O
II	Chamber sensed	A, V, D, O
III	Response to sensing	I (inhibit), T (trigger), D, O
IV	Rate modulation	R
V	Multisite	A, V, D

- **VVI:** paces/senses ventricle, inhibited by native beats
- **DDD:** physiological AV sequential pacing
- **AAI:** sinus node disease with intact AV conduction
- **VOO / DOO:** asynchronous (magnet, surgery with diathermy) – R-on-T risk

13. Pacemaker Problems

- **Failure to capture:** spike without QRS → ↑ output, reposition; hyperK, ischemia, lead displacement
- **Failure to pace (output):** no spike → oversensing, battery, lead fracture
- **Undersensing:** spikes inappropriately on native complexes → ↑ sensitivity (↓ mV value)
- **Oversensing:** pauses (T waves, myopotentials, EMI) → ↓ sensitivity
- **Pacemaker-mediated tachycardia** (DDD): retrograde P → magnet / ↑ PVARP
- **Magnet:** pacemaker → asynchronous; ICD → suspends tachy therapy (no pacing change)
- **Cardioversion with device:** pads ≥ 8 cm away, AP position; check device after

14. ICD & Electrical Therapy

- **ICD shocks:** appropriate (VT/VF) vs inappropriate (AF, T-wave oversensing, lead fracture)
- **Electrical storm:** ≥ 3 episodes / 24 h → sedation, amiodarone + β-blocker, correct K/Mg, ablation
- **Deactivate ICD** in end-of-life care (with patient / family consent)
- **Synchronized cardioversion:** AF 120–200 J biphasic; flutter / SVT 70–120 J; VT with pulse 120–150 J
- **Defibrillation:** VF / pulseless VT 150–200 J biphasic (360 J monophasic)

15. High-Yield MCQ Pearls

- [x] H^+ = 24 × $PaCO_2$ / HCO_3 checks ABG validity
- [x] Winter's: $PaCO_2$ = 1.5 × HCO_3 + 8 ± 2
- [x] Δ ratio > 2 = coexisting metabolic alkalosis
- [x] Osmolar gap > 10 → toxic alcohol
- [x] Paracetamol + malnutrition → pyroglutamic acidosis
- [x] Mobitz II → pacing; Mobitz I benign
- [x] Anterior MI block = infranodal (bad)
- [x] Atropine fails in heart transplant & infranodal block
- [x] TCA: R in aVR > 3 mm, QRS > 100 → bicarbonate
- [x] Magnet on ICD suspends shocks only
- [x] Transcutaneous capture: confirm by pulse
- [x] DDD = physiological AV pacing

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